A missing boundary in the Suzuki subfield construction

Human Compression project

Abstract

The number of distinct prime divisors of a field degree does not determine whether its smallest selected subfield is trivial. A repeated prime factor allows all distinct primes to be selected while leaving a simple Suzuki group at the bottom. This yields a rank-four residually weakly primitive pregeometry omitted from the eleven literal constructions in the stated classification. We give the complete subgroup argument, reusing the earlier project proof of this boundary. We do not assert a corrected exhaustive classification or a new solution of an open problem.

1 The subgroup condition

Let G be a group, let I be a finite type set, and let $G_i \le G$ for $i \in I$. The coset pregeometry has elements gG_i, of type i; two elements are incident when their cosets intersect. Put

$$G_J = \bigcap_{j \in J} G_j \quad (J \neq \varnothing), \qquad G_\varnothing = G.$$

The source calls this system residually weakly primitive when, for every $J \subsetneq I$, at least one $i \in I \setminus J$ makes $G_{J\cup\{i\}}$ a maximal proper subgroup of G_J. This is the definition on printed page 4 of the source. Its principal classification concerns pregeometries with precisely this subgroup condition; flag transitivity and the geometry axiom are not additional hypotheses of that classification.

Write $S_d = \mathrm{Sz}(2^d)$ for odd d. We use standard simultaneous subfield embeddings into the faithful four-dimensional Suzuki model. Two classical facts are retained: S_d is nonabelian simple for $d \ge 3$, and S_d is maximal in S_e if e/d is prime and $d > 1$. These are among the Suzuki structural inputs used by the source itself (its Table 1, printed page 6).

We also record why subfield intersections are exact in these embeddings. In characteristic two the standard root and diagonal matrices are

$$U(a,b) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ a & 1 & 0 & 0 \\ a^{\theta+1}+b & a^\theta & 1 & 0 \\ a^{\theta+2}+ab+b^\theta & b & a & 1 \end{pmatrix}, \qquad h(t) = \mathrm{diag}(t^{\theta+1}, t, t^{-1}, t^{-\theta-1}),$$

where $\theta = 2^{(e+1)/2}$ in $\mathbb{F}_{2^e}$; let w be the antidiagonal identity matrix. The standard Bruhat forms are $U(a,b)h(t)$ and $U(a,b)h(t)wU(c,f)$. This matrix description is retained classical Suzuki theory. For odd $d \mid e$, the exponent θ restricts to $2^{(d+1)/2}$ on $\mathbb{F}_{2^d}$.

Lemma 1. *For odd divisors a, b of an odd integer n, the standard embedded subgroups satisfy*

$$S_a \cap S_b = S_{\gcd(a,b)}.$$

Proof. First, the elements of S_n whose matrix entries lie in $\mathbb{F}_{2^d}$ are exactly S_d. In the first Bruhat form, the entry g_{22} gives t, the ratio g_{21}/g_{11} gives a, and g_{42}/t gives b. Thus all parameters lie in the smaller field when the entries do. In the second form,

$$g_{14} = t^{\theta+1}, \quad c = g_{13}/g_{14}, \quad f = g_{12}/g_{14}, \quad a = g_{24}/g_{14}, \quad b = g_{34}/g_{14} + a^{\theta+1}.$$

The power $\theta+1$ is coprime to 2^n-1: an odd common divisor would give both $2^{n+1}=1$ and $2^n=1$ modulo that divisor. Consequently t is a fixed integer power of g_{14}, and hence also lies in the smaller field. The two forms are distinguished by whether g_{14} is zero. This proves the asserted equality with the matrices over $\mathbb{F}_{2^d}$. Finally $\mathbb{F}_{2^a}\cap\mathbb{F}_{2^b}=\mathbb{F}_{2^{\gcd(a,b)}}$, which gives the result. □

2 The omitted pregeometry

Let $p_1,\dots,p_k$ be distinct prime divisors of an odd integer n, and put

$$d=\frac{n}{p_1\cdots p_k},\qquad G=S_n,\qquad G_i=S_{n/p_i}.$$

All embeddings are the standard ones just fixed.

Theorem 2. *If $d>1$, the resulting rank-k coset pregeometry is residually weakly primitive. In fact every available one-step inclusion in its intersection lattice is maximal, and all its 2^k intersections are distinct.*

Proof. Repeated use of Lemma 1 gives, for every $J\subseteq I$,

$$G_J=S_{n_J},\qquad n_J=\frac{n}{\prod_{j\in J}p_j}.$$

For $i\notin J$ the ratio $n_J/n_{J\cup\{i\}}$ is the prime p_i. The smaller exponent is at least $d>1$. Classical subfield maximality therefore applies to every such inclusion. Unique factorization makes the integers n_J distinct for distinct subsets J. The order $2^{2r}(2^{2r}+1)(2^r-1)$ of S_r is strictly increasing with r, so the corresponding subgroups are distinct as well. These statements give every condition in the source's definition. □

For an explicit example take

$$n=3^2\cdot5\cdot7\cdot11=3465,\qquad(p_1,p_2,p_3,p_4)=(3,5,7,11).$$

The four maximal parabolics are

$$S_{1155},\quad S_{693},\quad S_{495},\quad S_{315},$$

and their common intersection is $S_3=\mathrm{Sz}(8)$. Theorem 2 proves that this is a rank-four residually weakly primitive pregeometry.

Theorem 3. *The displayed pregeometry is absent from Constructions 1–11 in the fixed published edition of* The residually weakly primitive pre-geometries of the Suzuki simple groups. *It therefore contradicts the literal exhaustive assertion in its Theorem 5.*

Proof. Constructions 1–10 all include a soluble maximal parabolic: a dihedral group, an affine soluble group, a cyclic group extended by a group of order four, or the specified mixed root group. Every maximal parabolic in our example is nonabelian simple. Thus a permutation of types and an ambient group automorphism cannot identify our system with any of those constructions.

Construction 11 writes $n=p_1^{e_1}\cdots p_m^{e_m}$ with the p_i pairwise distinct, and permits ranks $\alpha\in\{2,\dots,m-1\}$. Here $m=4$, so its largest permitted rank is three. Our rank is four. Theorem 5 states that all residually weakly primitive pregeometries of rank at least four are those in Constructions 1–11, with no additional operation of taking truncations. The example is therefore outside that literal list. □

3 Attribution and limits

The source is D. Leemans, *The residually weakly primitive pre-geometries of the Suzuki simple groups*, Note di Matematica 20(1) (2000/2001), 1–20. The relevant definition is on printed pages 4–5, the construction conventions on page 15, and Construction 11 and Theorem 5 on page 18. The source was received on 26 July 1999 and accepted on 21 September 1999.

The argument is a reworking of Sections 2–3 of the project's earlier Suzuki subfield manuscript, whose separate review already checked this exact omitted boundary against the source. The old proof establishes additional geometry and flag-transitivity statements using descent. Those stronger conclusions are not needed here and are not acquired as premises of this note. The elementary matrix intersection argument comes from the earlier project treatment of compatible Suzuki tori. Classical Suzuki structure and prime-degree subfield maximality remain explicit retained inputs.

The correction concerns the literal rank range. The condition

$$n/(p_1 \cdots p_k) > 1$$

includes the example; suitable extensions of the construction can therefore repair this omission. We make no claim that such a repair exhausts all embeddings, and do not classify the remaining ten constructions. Neither a new mathematical discovery nor historical priority is claimed by this reworking. It contributes no additional conjecture solution, Lean theorem or compression factor. The supplementary control checks the matrix parameter identities symbolically, and the sixteen exponents and thirty-two prime-degree inclusions by integer arithmetic. It does not enumerate S_{3465}.